



**EMBEDDED SYSTEM FINAL PROJECT REPORT
DEPARTMENT OF ELECTRICAL ENGINEERING
UNIVERSITAS INDONESIA**

Automated Visitor Counter

GROUP 7

Mochammad Rafly Fatih Rabbani	2406369021
Derryl Liandryo Putra	2406368920
Ibrahima Sory SQUARE	2506698030

PREFACE

This project is formed in fulfilling the requirement of the Digital System Design Course Lab final project. This report documents the implementations during the creation of this project which includes theory, process and results of the project called the Automated Visitor Counter.

Depok, May 17, 2026

Group 7

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CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

Counting visitors traditionally becomes a problem as the number of visitors are always random and unpredictable, especially when it's tracked manually of how many visitors are in the area. It becomes a challenge to some places that require limiting the amount of visitors in specific times or events to maintain safety and order. Keeping track of the amount of visitors that are allowed entering the building or site becomes essential to reduce risks of overcrowding which potentially leads to environmental degradation, bad tourist experience, as well as safety during emergencies.

1.2 PROJECT DESCRIPTION

The Automated Visitor Counter is an integrated embedded system designed to monitor and record room attendance. This project provides a comprehensive solution for managing facility capacity, ensuring safety, and analyzing foot traffic patterns. By utilizing instantaneous sensor detection, real-time data processing, and memory logging, the project delivers a highly responsive and reliable counting management tool.

1.3 OBJECTIVES

The objectives of this project is to design and implement modules 2,3,4,5,6,7, and 9 to develop the Automated Visitor Counter as a medium to demonstrate our understanding.

1.4 ROLES AND RESPONSIBILITIES

The roles and responsibilities assigned to the group members are as follows:

Roles	Responsibilities	Person
visitor_counter.s	The main file that will integrate the whole module functionalities from the three functions that were created.	Everyone
core_interrupt.S	Configure entry and exit IR sensors with external interrupts and flags.	Derryl
display_timer.S	Configure Timer0 overflow interrupt, LCD display refresh, green/red LED indicators, and buzzer control based on the occupancy limit status.	Ibrahima
data_communications.S	Configure occupancy counter operations, EEPROM logging, RTC timestamp retrieval via I2C, and USART communications.	Fatih

Table 1. Roles and Responsibilities

CHAPTER 2

IMPLEMENTATION

2.1 EQUIPMENT

The tools that are going to be used in this project are as follows:

- Arduino (IDE, Electrical Components)
- Proteus
- Google Docs
- Github, Git

2.2 IMPLEMENTATION

2.2.1 Module 2 - I/O

The I/O module implements the input system of the Automated Visitor Counter using infrared (IR) sensors connected to the external interrupt pins of the ATmega328P microcontroller. Two IR sensors are used in our system: one sensor is assigned for entry detection and another for exit detection. The sensors are connected to the INT0 and INT1 interrupt pins (Pins 2 and 3) to allow immediate response whenever a visitor enters or exits the monitored area.

The external interrupt mechanism is used to detect signal changes generated by the IR sensors which are configured on FALLING EDGE. When the entry sensor is triggered, our corresponding ISR sets the **ENTRY_FLAG** variable. Similarly, when the exit sensor is triggered, the ISR sets the **EXIT_FLAG** variable. We keep our interrupt routines minimal and lightweight to ensure fast execution and system stability. Complex operations such as updating the occupancy count, EEPROM logging, I2C communication, USART transmission, and LCD refresh are not handled inside our ISR.

This module allows our system to perform real-time event detection efficiently without continuously polling the sensors. By using hardware interrupts, the microcontroller can react immediately to visitor movement while maintaining low processing overhead and reliable operation.

2.2.2 Module 3 - USART

We used the USART module to implement serial communication between the ATmega328P microcontroller and the serial monitoring interface through the UART communication protocol. The TX and RX pins of the microcontroller are used to transmit visitor activity data to the Proteus Virtual Terminal during simulation and testing. The USART peripheral is configured with a baud rate of 9600 bps by setting the `UBRR0L` register to 103.

Whenever a visitor enters or exits the monitored area, we send a clear message to the screen through the USART interface. The transmitted data includes the event type, such as ENTRY or EXIT, along with the timestamp retrieved from the RTC module. This communication mechanism allows our system to provide real-time monitoring and debugging capabilities during operation.

It is important for us to highlight that the USART module operates together with the occupancy management system and supports continuous transmission without blocking the main execution flow of the program. That's how our project can store, visualize, and analyze visitor activity externally while maintaining efficient embedded system performance.

2.2.3 Module 4 - Arithmetic

We used the Arithmetic module to process the input flags generated by our external interrupt module and perform arithmetic operations to maintain the current number of visitors inside the monitored area.

When the `ENTRY_FLAG` is activated by the entry IR sensor, our system increments the `OCCUPANCY_COUNT` variable. Conversely, when the `EXIT_FLAG` is activated by the exit IR sensor, our system decrements the occupancy value. Arithmetic operations are performed carefully to ensure that the occupancy count never becomes negative and does not exceed the predefined maximum capacity limit of 5 visitors.

With this module, we also compare the current occupancy value with the maximum allowed capacity. If the occupancy reaches or exceeds the configured limit of 5 visitors, the

`ALARM_ACTIVE` flag is enabled and later used by the display and alarm module to activate the buzzer and control the relay system.

2.2.4 Module 5 - Timer

We used Timer0 for generating regular interrupts used for controlling time-dependent system outputs such as the buzzer and LED indicators. Timer0 is configured in normal mode with a prescaler value of 64 selected to generate overflow interrupts approximately every 1.024 ms.

Each timer overflow triggers an Interrupt Service Routine (ISR) that continuously checks the `ALARM_ACTIVE` flag. When the occupancy limit is reached, the timer-controlled logic activates the red LED and buzzer while turning off the green LED to indicate an alert condition. When the occupancy returns below the limit, the buzzer is turned off, the red LED is disabled, and the green LED returns to normal operation.

This timer-based implementation allows our system to react continuously to occupancy changes without using blocking software delays, improving the responsiveness and stability of the project.

2.2.5 Module 6 - Interrupt

As we mentioned before, this module enables our system to detect visitor movement immediately whenever the infrared entry or exit sensors are triggered.

The entry IR sensor is connected to the INT0 pin 2, while the exit IR sensor is connected to the INT1 pin 3. Both interrupts are configured to respond to FALLING EDGE signal changes generated by the sensors. When an interrupt occurs, the corresponding Interrupt Service Routine (ISR) is executed automatically by the microcontroller.

The interrupt routines are designed to remain short and efficient. Instead of performing complex operations directly inside the ISR, the routines only update the event flags `ENTRY_FLAG` and `EXIT_FLAG`. The interrupts are also temporarily disabled inside the ISR to prevent repeated triggering while the sensor remains active.

These flags are then processed by our arithmetic and communication modules to update the occupancy count, trigger alarms, store logs in EEPROM, refresh the LCD display, and transmit serial data as mentioned previously.

2.2.6 Module 7 - EEPROM

With the EEPROM module we implement non-volatile data storage using the internal EEPROM memory of the ATmega328P microcontroller. We used this module to preserve important system data even when the power supply is turned off or the microcontroller is restarted.

In our system, the EEPROM is used to store the latest occupancy count and selected visitor activity records. Whenever the occupancy value changes due to an entry or exit event, the updated count is written into EEPROM memory sequentially using the `EEP_PTR` pointer variable. This ensures that the system can recover the previous occupancy state after a reset or power interruption.

The EEPROM write and read operations are handled carefully to maintain data integrity and reduce unnecessary memory wear. Before every EEPROM access, our system checks whether the EEPROM is busy to avoid corrupted operations. EEPROM access is performed only when necessary because EEPROM memory has a limited number of write cycles.

When our system starts, the stored occupancy value is read back from EEPROM and restored into the `OCCUPANCY_COUNT` variable. This allows our counter to continue operating without losing previously recorded data.

2.2.7 Module 9 - I2C

In the I2C module we implement communication between the ATmega328P microcontroller and the Real-Time Clock (RTC) module using the Inter-Integrated Circuit (I2C) protocol. This communication allows our system to retrieve accurate time information for visitor activity logging.

The ATmega328P operates as the I2C master device, while the DS3231 RTC module operates as the slave device. The SDA (Serial Data) and SCL (Serial Clock) lines are used to exchange data between our two components. Through this communication, the system can read the current hour, minute, and second values from the RTC module whenever a visitor entry or exit event occurs.

The timestamp information obtained from the RTC is combined with the occupancy data and transmitted through the USART module for real-time monitoring and logging purposes.

In fact, we selected the I2C protocol because it allows efficient communication using only two signal lines while supporting reliable real-time data exchange. By integrating the RTC module through I2C communication, our system can maintain accurate visitor event timestamps independently from the microcontroller execution timing.

CHAPTER 3

TESTING AND ANALYSIS

3.1 TESTING

The testing phase of our Automated Visitor Counter project was carried out using both Proteus simulation and hardware-level verification through the ATmega328P microcontroller environment. Each module was tested individually before integrating the complete system in order to ensure stable operation and proper communication between all components.

First, we tested the external interrupt module by triggering the entry and exit IR sensors connected to pins INT0 and INT1. During this test, we verified that the corresponding `ENTRY_FLAG` and `EXIT_FLAG` variables were correctly activated whenever a visitor crossed the sensors. We also confirmed that the interrupt mechanism reacted immediately without requiring continuous polling from the microcontroller.

Next, we tested the arithmetic processing module by simulating multiple entry and exit events. During this stage, we monitored the `OCCUPANCY_COUNT` variable to ensure that the counter incremented correctly when a visitor entered and decremented correctly when a visitor exited. Additional testing was performed to verify that the occupancy value never became negative and that the alarm condition was activated once the occupancy limit of 5 visitors was reached.

The USART communication module was tested using the Proteus Virtual Terminal. We verified that the ATmega328P transmitted real-time serial messages containing the visitor event type and RTC timestamp. The transmitted messages were displayed correctly through UART communication at 9600 baud.

The I2C communication module was also tested with the DS3231 RTC module. During testing, we verified that the system successfully retrieved the current hour, minute, and second values through the SDA and SCL communication lines. The RTC timestamps were then combined with the USART messages and EEPROM logs.

The EEPROM module was tested by storing occupancy values and visitor logs into the internal EEPROM memory of the ATmega328P. After resetting the system, we verified that

the previously stored occupancy value was successfully restored from EEPROM memory, confirming proper non-volatile storage functionality.

Finally, the timer and alarm module were tested by simulating occupancy values below and above the predefined limit. When the occupancy remained below the limit, the green LED stayed active while the buzzer and red LED remained disabled. Once the occupancy reached the limit of 5 visitors, the red LED and buzzer were activated correctly, demonstrating proper alarm handling through the Timer0 interrupt routine.

3.2 RESULT

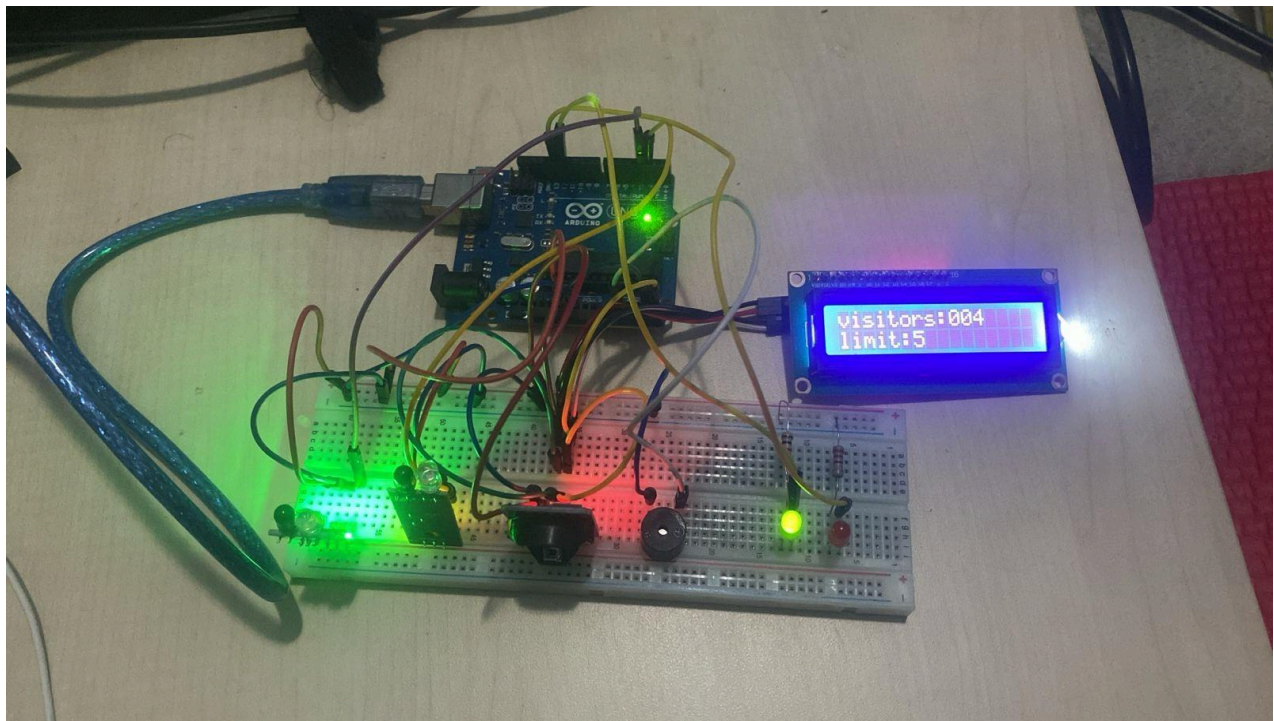


Fig. 1. Normal operating condition when visitors = 004 and limit = 5.

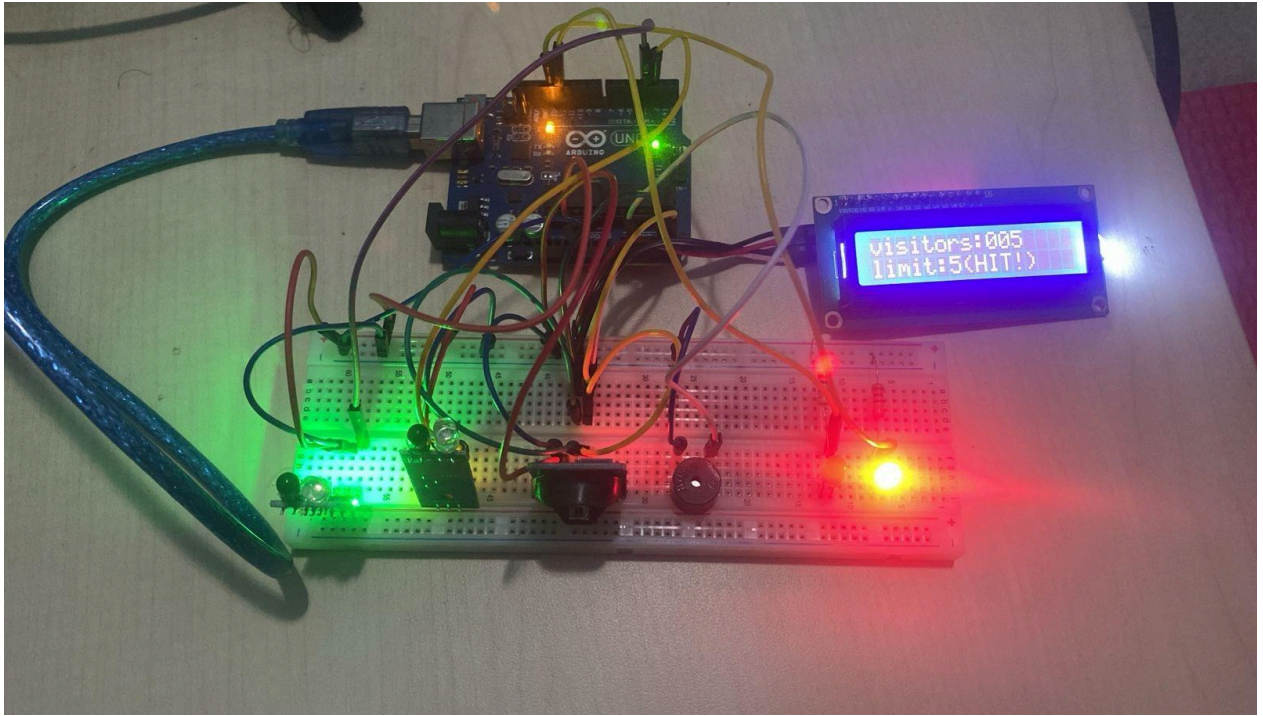


Fig. 2. Alarm condition: occupancy reaches the limit of 5 visitors.

3.3 ANALYSIS

Based on the testing results, our Automated Visitor Counter system was able to operate successfully in both normal and alarm conditions. All implemented modules communicated correctly through the ATmega328P microcontroller and produced stable real-time responses during operation.

From the hardware testing, we observed that the infrared sensors correctly detected visitor entry and exit events. Whenever a visitor crossed the entry sensor, the occupancy count increased automatically. Similarly, when the exit sensor was triggered, the occupancy count decreased correctly. The use of external interrupts allowed the system to react immediately without continuously polling the sensor inputs, which improved the responsiveness and efficiency of the system.

The LCD display module also operated correctly during testing. The current occupancy value was refreshed in real time and displayed together with the occupancy limit information.

When the occupancy remained below the maximum limit of 5 visitors, the system stayed in normal operating mode where the green LED remained active and the buzzer stayed disabled.

When the occupancy count reached the maximum limit, the `ALARM_ACTIVE` flag was triggered successfully. During this condition, the red LED and buzzer were activated while the green LED was turned off. The LCD displayed the message “limit:5(HIT!)”, clearly indicating that the maximum capacity had been reached. This demonstrates that the arithmetic and timer modules interacted correctly with the alarm system.

The USART communication module also produced stable results. Visitor activity logs containing ENTRY and EXIT events together with RTC timestamps were successfully transmitted through UART communication and displayed on the Proteus Virtual Terminal. In addition, the I2C communication with the DS3231 RTC module successfully retrieved hour, minute, and second information for accurate event logging.

The EEPROM module functioned correctly as well. During testing, the occupancy value was successfully stored into EEPROM memory and restored after restarting the system. This confirmed that the system was capable of preserving important data even after power interruption or reset conditions.

Overall, the obtained results demonstrate that our Automated Visitor Counter system achieved its main objectives successfully. The system was capable of performing reliable real-time visitor monitoring, occupancy counting, alarm handling, serial communication, RTC timestamp retrieval, LCD display control, and EEPROM data storage within a single embedded system implementation.

CHAPTER 4

CONCLUSION

In conclusion, our Automated Visitor Counter project was successfully designed and implemented using the ATmega328P microcontroller together with several embedded system modules such as external interrupts, USART communication, arithmetic processing, Timer0 interrupts, EEPROM memory, and I2C communication with the DS3231 RTC module.

Through this project, we were able to develop a system capable of detecting visitor entry and exit events in real time using infrared sensors connected to external interrupt pins. The occupancy count was updated automatically and displayed continuously on the LCD screen. In addition, the system successfully handled occupancy limit monitoring by activating alarm indicators such as the buzzer and red LED whenever the maximum capacity of 5 visitors was reached.

The USART module allowed us to monitor visitor activity logs and timestamps through serial communication, while the EEPROM module ensured that occupancy data could still be preserved after restarting the system. Furthermore, the I2C communication with the RTC module enabled accurate timestamp retrieval for event logging purposes.

Overall, this project helped us improve our understanding of embedded system concepts already initiated in class and lab including hardware interrupts, timer configuration, UART communication, EEPROM memory management, I2C communication, and real-time system integration. The final implementation demonstrated that multiple embedded modules can work together efficiently to create a reliable real-time monitoring system.

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APPENDICES

Appendix A: Project Schematic

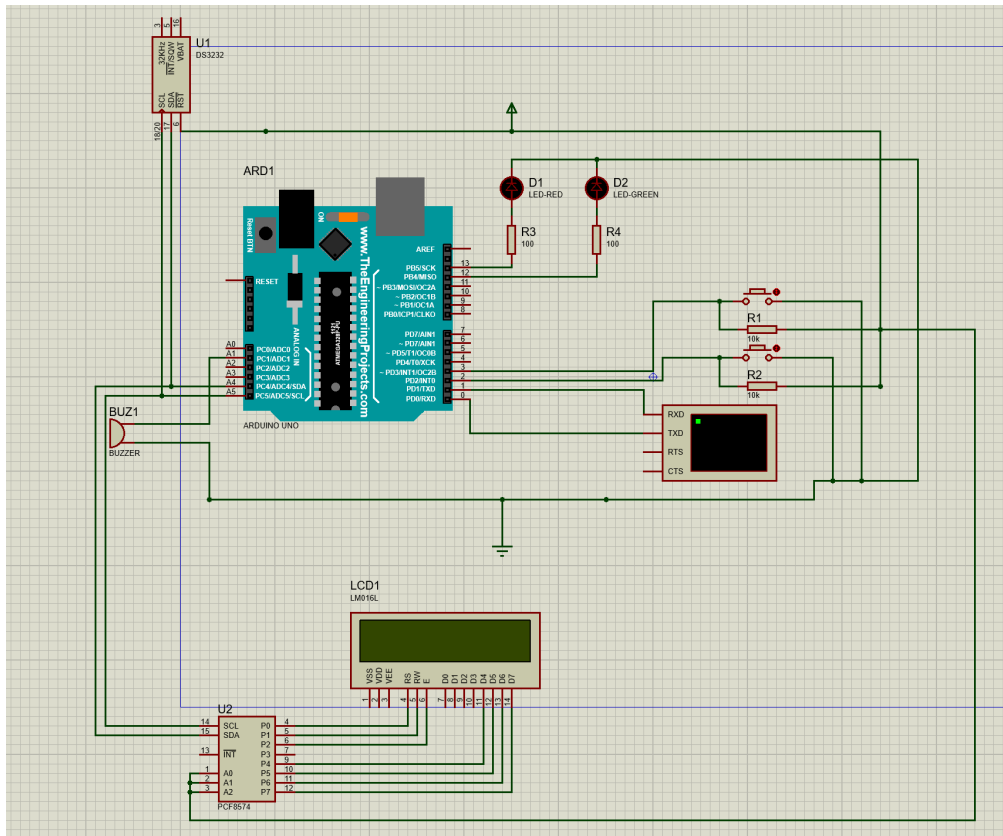


Fig 3. Schematic

We used buttons because the IR sensor model does not work on proteus.

Appendix B: Documentation

